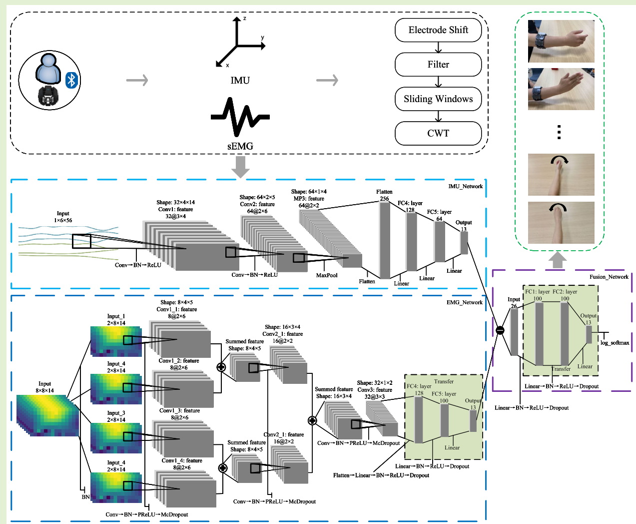
sEMG and IMU Data-Based Hand Gesture Recognition Method Using Multistream CNN With a Fine-Tuning Transfer Framework

Guiyin Li, Bo Wan, Kejia Su, Jiwang Huo, Changhua Jiang, and Fei Wang



***Abstract*—Interfaces based on surface electromyography (sEMG) and inertial measurement units (IMUs) enable users to interact with computers in a natural and intuitive way through hand gestures. sEMG or IMU sensor-based approaches can provide muscle electrical activity or motion information to recognize gestures. However, the existing methods recognize static and dynamic gestures separately and hierarchically. In addition, the variability in sEMG data among different subjects limits the performance of sensor-based human–computer interfaces. To address these limitations, this article proposes a multistream convolutional neural network (CNN) architecture with a fine-tuning transfer strategy. The multistream architecture explores the complementary nature of sEMG and IMU signals and achieves real-time recognition of dynamic and static gestures in a nonhierarchical way. In addition, a new dataset including seven static gestures and six dynamic gestures is designed for evaluation. The proposed method is verified by experiments on six subjects and is compared with previous methods on the same dataset. Experimental results reveal that the transfer strategy significantly improves recognition accuracy from 89.99% to 98.49%. Meanwhile, by learning the fusion features of sEMG and IMU signals, the recognition accuracy is enhanced from 49.54% (sEMG) and 59.91% (IMU) to 89.99%. In addition, the accuracy and latency of this method for real-time recognition are 98.12% and 103 ms, respectively. These results demonstrate that the proposed multistream CNN model accurately recognizes these static and dynamic gestures online in a nonhierarchical way and can effectively utilize the complementarity between sEMG and IMU signals.**

***Index Terms*— Convolutional neural network (CNN), hand gesture recognition (HGR), inertial measurement unit (IMU), multistream fusion, surface electromyography (sEMG), transfer learning (TL).**

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I. Introduction

IN THE field of human–computer interaction (HCI), hand gestures enable users to communicate with external devices. Hand gesture recognition (HGR) has been widely employed in HCI applications such as robot control [1] and virtual reality [2]. However, maintaining the high accuracy and robustness of HGR has remained a challenging task.

Generally, there are two types of HGR methods. The first is computer vision-based recognition methods, which use devices with cameras (e.g., Kinect sensors [3] and Leap Motion sensors [4]) to collect images of hand gestures. Sun et al. [3] collected the RGB-D images of static gestures using a Kinect sensor and proposed a two-stream feature fusion strategy. Experiments showed that the average accuracy of their proposed algorithm was 1.08% higher than that of the single-stream model. Li et al. [4] captured images of hand gestures using Leap Motion sensors and proposed an initial

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frame correction strategy to quickly initialize the trajectory of test gestures. The recognition accuracy rate of this method was higher than 90%. Moreover, Lu et al. [5] captured images of dynamic hand gestures using Leap Motion sensors and then constructed the LeapMotion-Gesture3D dataset. A feature vector was computed and fed to a hidden conditional neural field classifier, and a classification accuracy of 89.5% was achieved. Although various computer vision-based approaches have achieved high classification accuracy, illumination changes and occlusion have particular impacts on recognition accuracy.

The second type of HGR method includes sensor-based approaches. These methods acquire hand gesture data using various wearable sensors providing different types of data. For instance, surface electromyography (sEMG), which is measured by multiple electrode sensors, reflects the electrical signal of muscle activity directly. Turlapaty et al. [6] proposed a classification framework to classify features extracted from multichannel sEMG signals. Inertial measurement units (IMUs), including accelerometers and gyroscopes, have the ability to recognize the movement of human arms. Recent studies have shown that using the fusion of sEMG and IMU data can improve classification accuracy and characterize complex interaction activities [7], [8], [9]. Chang et al. [9] developed a hierarchical strategy to recognize six hand motions based on data collected by sEMG and IMU sensors. According to the judgment result of rotary motions, sEMG and IMU signals were used separately to classify hand motions. Jiang et al. [8] introduced a real-time gesture recognition method based on the fusion of IMU and sEMG signals. The time-domain features of the sEMG and IMU were extracted. After extracting features, the linear discriminant analysis (LDA) method was used to recognize gestures. Pan et al. [10] proposed a hierarchical recognition method to identify significant and subtle motion gestures from sEMG and ACC data. Then, handcrafted features were combined with deep belief network (DBN) features to obtain the recognition result.

The previously proposed methods mainly extract handcrafted features or perform classification based on the sEMG and IMU signals independently. By contrast, hierarchical recognition frameworks have been mainly proposed to recognize static gestures or dynamic gestures. These frameworks first classify the input signals into a static gesture or a dynamic gesture. Then, two classifiers (i.e., the static gesture classifier and the dynamic gesture classifier) are trained separately to identify gestures. However, these approaches are time-consuming and cumbersome. To address these shortcomings, the present study uses the Myo armband, which contains embedded sEMG and IMU sensors, to obtain muscle activation and movement information on gestures.

Furthermore, this work focuses on adopting deep learning-based approaches to decode the sEMG and IMU signals. Compared with machine learning-based methods, deep learning-based approaches can extract features automatically from a large quantity of input data while achieving higher gesture recognition accuracy [11], [12], [13], [14]. Zia et al. [13] input raw EMG signals into a proposed convolutional neural network (CNN). Experimental results demonstrated that deep learning-based techniques outperformed machine learning-based algorithms. Chen et al. [14] proposed a 3-D CNN for HD-EMG-based gesture classification, and their results showed that the 3-D convolution network could achieve better performance than the combination of 2-D convolution and majority voting. Although many gesture recognition tasks have been successfully addressed using deep learning methods, the accuracy of gesture recognition is still affected by the quantity of available data, sensor position, and differences across subjects. Large quantities of training data could be obtained by aggregating signal data from multiple participants. However, this might bring a significant training burden to users. Considering this problem, Castellini et al. [15] stated that EMG signals differ significantly between subjects despite adjusting electrode locations. Recently, transfer learning (TL) has been proposed and widely used to overcome the aforementioned shortcomings [16], [17], [18]. In the present study, we combine a deep learning-based algorithm and a TL strategy.

In this work, a multistream CNN model with a fine-tuning transfer strategy is proposed to recognize static and dynamic gestures in a non-hierarchical way. This refers to the approach where gestures are directly classified without distinguishing between dynamic and static gestures. Compared with hierarchical methods, this approach has the potential to recognize combined dynamic and static gestures in the future. This is one of the motivations for studying a non-hierarchical approach. Another motivation is to perform feature fusion of sEMG and IMU to learn more intrinsic information. The main contributions of our work are as follows. First, we constructed a hand gesture dataset containing 13 types of gestures, including seven static gestures and six dynamic gestures, collected from 21 subjects. The gesture data used in this study were captured by a Myo armband, which contains embedded IMU and sEMG sensors. To the best of the authors' knowledge, this is the first Myo-dataset containing both static and dynamic gestures sensed by eight-channel sEMG signals and six-channel IMU signals, including three-axis acceleration and three-axis gyroscope signals. Our second contribution is that a multistream CNN with a fine-tuning transfer framework was proposed to recognize hand gestures across subjects. First, the eight-channel sEMG signals were divided into four streams, which were then sequentially fused. Afterward, the fused sEMG signals were further combined with the IMU. By learning the features before and after the fusion of multiple muscle groups and motion information, a source domain model was trained. Subsequently, fine-tuning was performed to transfer the model to the target domain. The third contribution is that the proposed framework represents an efficient nonhierarchical method for classifying static and dynamic gestures by fusing sEMG and IMU signals. Moreover, the proposed framework with the transfer strategy performed accurate real-time classification of gestures from new users.

The remainder of this article is organized as follows. Section II describes the proposed framework in detail. In Section III, the experimental results and a discussion are presented. Section IV provides validation for real-time recognition. Finally, conclusions and directions for future works are given in Section V.

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II. Proposed Method

A. Data Acquisition and Preprocessing

There were 21 non-disabled participants in this study. Approval of the experiment was granted by the Research Ethics Committee of Xidian University. Before the experiment, each participant was informed of the experimental process and protocols, and all participants signed an informed consent form. An LED screen was used to display the gesture instruction video. During experiments, participants were asked to sit in a comfortable position in a quiet room. A Myo armband was used to capture gesture activity. Once wearing the Myo armband, each participant was first required to perform a hardware calibration procedure: straightening the five fingers together and bending the wrist outward by 90° until the MYO armband ceases vibrating. Subsequently, the participants executed 13 types of gestures following the given instructions, as shown in Fig. 1. The eight-channel sEMG signals were collected at a 200 Hz sampling rate. Meanwhile, the three-channel acceleration signals (Acc) and three-channel gyroscope signals (Gyro) were captured at a 50 Hz sampling rate.

During data acquisition, each type of gesture was executed in six repetitive trials. There was a 5 s rest between trials to prevent muscle fatigue. The raw signal data of the 21 subjects included 1638 ($13 \times 21 \times 6$) gestures, and a dataset named the XDMyo[1] dataset was constructed. The XDMyo dataset contained seven static gestures and six dynamic gestures. Participants performed each static gesture within 5 s and each dynamic gesture within 3 s. The XDMyo dataset was partitioned into a source dataset and target dataset. The source dataset comprised data from 15 participants. Through parameter tuning, we determined the highest accuracy of the source domain model by splitting the source dataset into training and validation sets using fourfold cross-validation. The target dataset consists of the remaining six participants. To fully leverage the data from these six individuals, we employed leave-one-out cross-validation to select an evaluation set. Each participant's data were repeated six times. Five repetitions were divided into training and validation sets using ten-fold cross-validation for fine-tuning the source domain model. The remaining repetitions were used as an evaluation set. As for the source domain model, all trials from these six individuals were used as an evaluation set.

After data acquisition, we conducted preprocessing. The effective frequency range of sEMG signals is from 5 to 500 Hz [19]. Although the sampling frequency of the Myo device's EMG signals is limited to 200 Hz, it is capable of capturing a significant portion of the effective signal range. To remove low-frequency noise below 5 Hz, a second-order Butterworth filter with a cutoff frequency of 5 Hz was employed to process the EMG signals. However, due to the subsequent feature extraction stage utilizing the Mexican Hat wavelet function in the continuous wavelet transform (CWT) and the Hann window function in the short-time Fourier transform (STFT), which inherently act as low-pass filters, we did not apply additional low-pass filtering of the EMG signals during the preprocessing stage. To have a consistent sampling frequency, the IMU signals (including Gyro and Acc signals) were upsampled to 200 Hz. Subsequently, the three-axis acceleration of dynamic hand gestures was normalized; this was followed by smoothing the IMU signals using a third-order Gaussian filter. The IMU signals before and after filtering are shown in Fig. 2. Finally, to satisfy the requirements for real-time gesture interaction, multiple window sizes within 300 ms were explored (i.e., 200, 230, 260, and 280 ms) [20], [21]. After evaluation, we found that the best performance was achieved with a sliding window size of 280 ms and a stride of 100 ms. Therefore, the signals were preprocessed using a sliding window as the smallest processing unit.

[1] https://github.com/GuiyinLi/XDMyo-Dataset.git

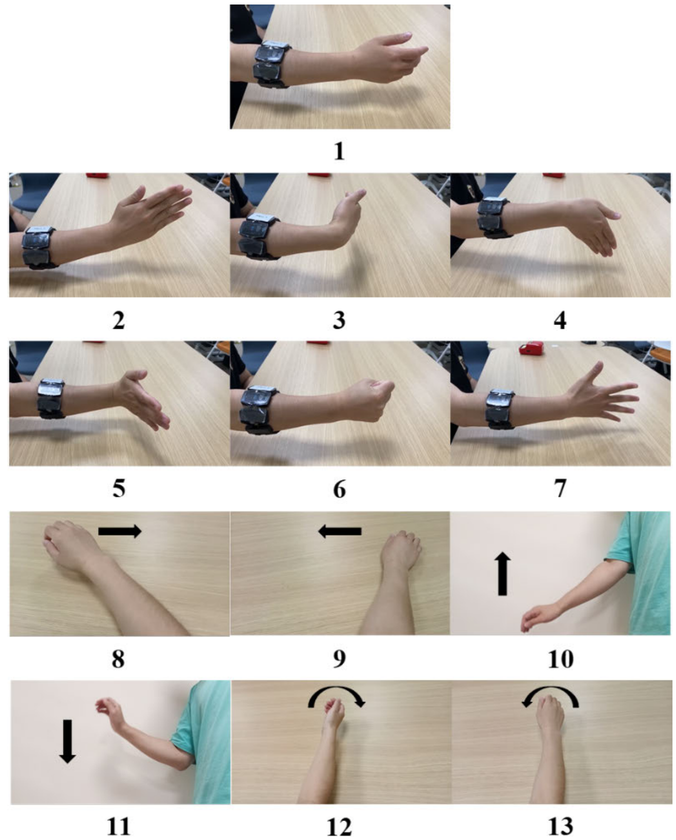

Fig. 1. Hand gestures in the XDMyo dataset. The arrows indicate the motion direction of dynamic gestures.

B. Feature Extraction

Due to the nonstationary nature of the sEMG signal, extracting features solely in the time or frequency domain is insufficient to capture meaningful information. Therefore, we compared two time-frequency domain feature extraction methods: the STFT and CWT.

In the STFT approach, the individual channel data (1×56) was segmented into multiple Hann windows, and then the Fourier transform was applied to each window. The window size was 16 sample points and the overlap size was 13 sample points. This process generated data with dimensions of 9×14. Subsequently, low-frequency data were removed from the spectrogram, resulting in final feature sample dimensions of $8 \times 8 \times 14$.

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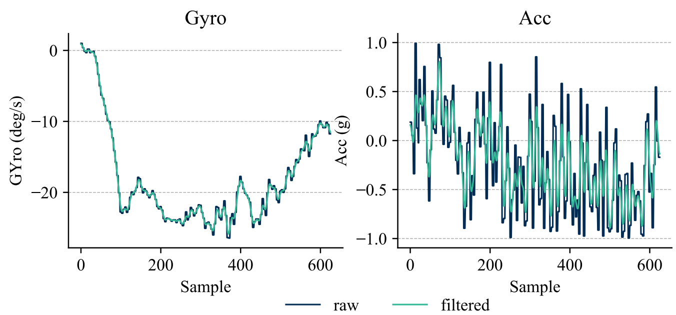

Fig. 2. Comparison of IMU signals before and after filtering, taking the Z-axis as an example.

In the CWT approach, the Mexican Hat function was used to extract the time-frequency domain features for each channel. With a sampling rate of 200 Hz, the wavelet transform scale range was set from 1 to 33 to mitigate the influence of low-frequency irrelevant information, thus giving rise to a feature matrix with dimensions of 32×56. Consequently, a spline interpolation of order 0 with a scaling factor of 0.25 was employed in the feature matrix for dimensionality reduction. This reduction process yielded an $8 \times 8 \times 14$ feature sample.

In addition, we utilized the raw data as the input of the proposed network for recognition. However, due to the mismatch in dimensionality between each sample point of the raw data (8×56) and the fusion network, an expansion process was performed to augment the dimensionality of the raw data to $8 \times 1 \times 56$. Furthermore, to ensure that the data has the same dimension as the input of the first fully connected layer in the EMG_Network branch, both the convolution kernel sizes and strides within the network (illustrated in Fig. 3) were modified.

C. Proposed Gesture Recognition Framework

After sEMG and IMU data collection and preprocessing, the signals were input into the proposed gesture recognition framework, as shown in Fig. 3. The framework mainly consisted of three branches: IMU_Network, EMG_Network, and Fusion_Network.

In the IMU_Network branch, the original dimension of the IMU data was 6×56 (i.e., Channel $\times$ Sample), where 6 represents the 3-D acceleration and angular velocity along the x, y, and z axes, and 56 represents the length of a sliding window. To use such data as input for the two-dimensional (2-D) CNN, we expanded the original data to $1 \times 6 \times 56$. The first two layers were 2-D convolutional layers consisting of 32 filters of size 3×4 with a stride of 1×4, and 64 filters of size 2×6 with a stride of 2×2. Batch normalization and Rectified Linear Unit (ReLU) nonlinearity functions were applied after each convolutional layer. These two convolutional layers were followed by a MaxPool layer, which consisted of 64 filters of size 2×2 with a stride 2×1. The output of the pooling layers was flattened and fed into fully connected layers. The last three layers were all fully connected layers with 128, 64, and 13 units.

In the EMG_Network branch, the original dimension of the EMG signal is 8×56 (i.e., Channel $\times$ Sample). In this study, both the CWT and STFT were used to extract the time-frequency domain features for each channel. The extracted features have a dimension of $8 \times 8 \times 14$ (i.e., Channel $\times$ Scale $\times$ Time for the CWT and Channel $\times$ Frequency $\times$ Time for the STFT). Since one or two channels can cover a specific muscle group in the upper limb, indicating a correlation between channels [22], [23], the eight-channel input was separated into four branches, each of which consisted of two channels. These four branches were first fed into the first 2-D convolutional layer, which consists of eight filters of size 2×6 with a stride of 2×2. The output data were then fused pairwise and input into the second convolutional layer, which consists of 16 filters of size 2×2 and stride 1×1. Finally, the fused data was passed to the third convolutional layer, which consists of 32 convolutional kernels of size 3×3 and stride 1×1. Finally, the data were fused again and fed into the third convolutional layer, which consists of 32 filters of size 3×3 and stride 1×1. The final three layers were fully connected layers with 128, 100, and 13 units. Due to the nonstationarity of sEMG [24], differences in signals generated by different users performing the same gestures can cause overfitting. By contrast, the IMU is a physical signal and does not exhibit this issue, and hence a Dropout layer is not necessary. Since Dropout is not suitable for convolutional layers, batch normalization, Parametric Rectified Linear Unit (PReLU) [25], and McDropout [26] are used after each convolutional layer instead. In the subsequent fully connected layers and the fusion network, we do use both ReLU and Dropout layers. While training the source domain model, both McDropout and Dropout have a rate of 0.1, which is changed to 0.2 during fine-tuning of the network model.

In the Fusion_Network branch, we employed a middle fusion strategy. The output vectors of the IMU_Network and EMG_Network branches were concatenated, and the resulting vector was fed into the Fusion_Network branch. Fusion_Network primarily consists of three fully connected layers with 100, 100, and 13 units. The batch normalization, ReLU, and Dropout were applied after each fully connected layer to prevent overfitting. The LogSoftmax activation function was used to transform the original network output into a logarithmic probability distribution. Meanwhile, the negative log-likelihood loss (NLLLoss) was utilized as the loss function to compute the loss between predicted labels and true labels.

D. Two-Stage Multistream Fusion Strategy

Multistream fusion has been widely used in image and video classification tasks. The fusion of multiple streams can provide complementary information and increase the accuracy of decision making. The principle behind a multistream fusion strategy is to select different data, construct neural network branches or extract features separately, and finally fuse and classify the multistream data. In this study, a two-stage multistream fusion strategy is used.

The fusion strategy based on the eight-channel sEMG signals was applied in the first stage of fusion. The Myo armband used eight sEMG sensors equally distributed around the forearm to measure muscle activity. When participants performed hand gestures, different forearm muscles contracted to different degrees; we also noted that forearm muscles were unevenly distributed. Learning correlations between specific muscles and gestures only by analyzing single-channel

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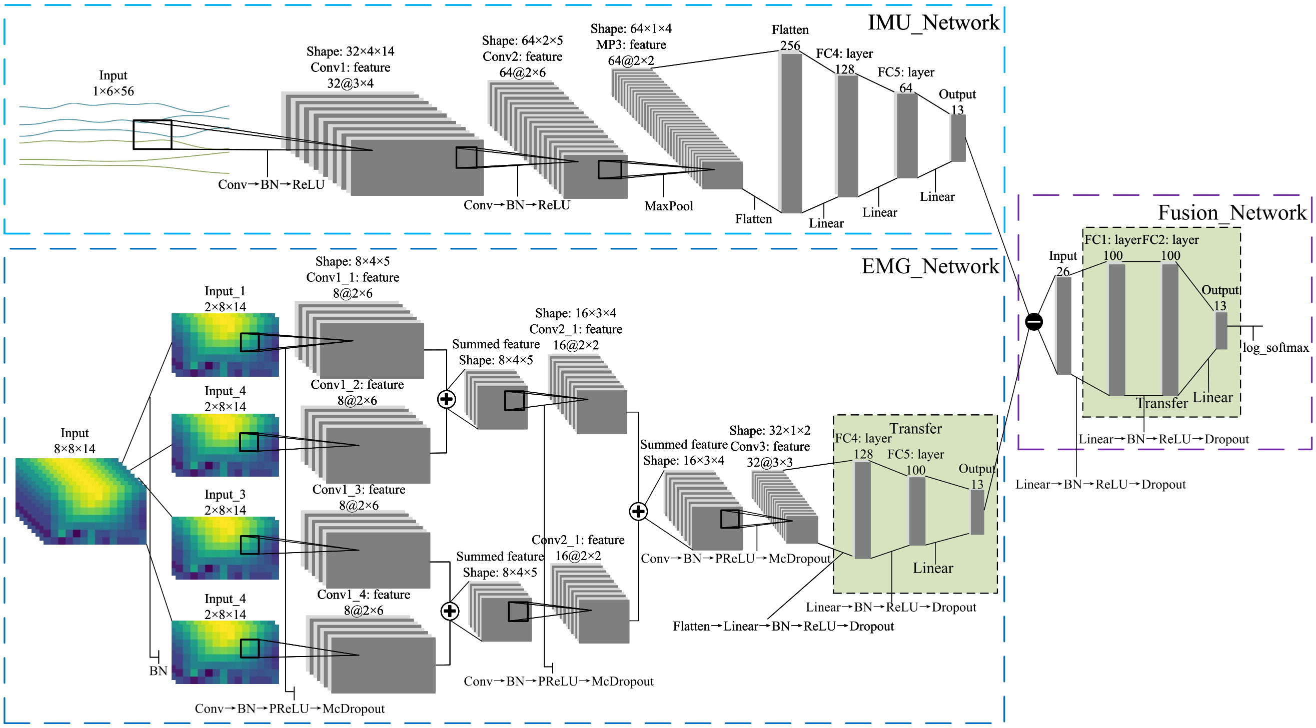

Fig. 3. Proposed multistream CNN model with a fine-tuning transfer framework. In this network architecture, the fully connected layer is denoted as FC, channel fusion is represented by "+," signal fusion is denoted by "−," the number before "@" indicates the number of filters, and the number after "@" represents the kernel size. The IMU_Network branch is enclosed in a dashed cyan box, the EMG_Network branch is enclosed in a dashed blue box, the Fusion_Network branch is enclosed in a dashed purple box, and the shaded green area represents the fine-tuning part of TL.

data has certain limitations. Therefore, in this work, the eight-channel sEMG signals were separated into four branches, each of which included two channels. The four branches were regarded as four streams and input into convolutional layers to learn the correlation between different muscles and specific gestures. Then, the four streams were fused into dual streams, which were further input into convolutional layers to continue learning relevant features. Finally, the outputs of the dual streams were fused to obtain the summed features of sEMG signals. This fusion process involved gradually combining different channels in pairs (i.e., employing a slow-fusion architecture [27]); these pairs are indicated by the "+" signs in the EMG_Network branch of Fig. 3.

In the second stage of fusion, the sEMG signals and IMU signals were regarded as two streams, and they were input into two separate networks to extract features. The extracted features were fused by applying the middle fusion strategy to complement the features of the sEMG and IMU data; the "-" sign in the Fusion_Network section of Fig. 3 denotes this process. During the fusion process, the feature vectors of both sEMG signals and IMU signals were concatenated together while preserving their original contents.

E. TL Strategy

When the sEMG signals and IMU signals were used for gesture recognition, there arose two main problems that impacted recognition accuracy. The first was electrode shift caused by variations in their placement. Although the Myo device undergoes hardware calibration after being worn, it could not guarantee consistent placement for each wearing session. In addition, the sensors inevitably showed certain lateral or longitudinal offsets over time, resulting in a misalignment of the eight channels of the sEMG signals. The second problem was decreased recognition accuracy due to interuser variations. Since the sEMG signals are biological signals, the motor potential generated by muscle activity varies due to individual differences among participants [15]. However, IMU signals, being physical signals, do not exhibit such variations. Therefore, it was observed that if the subject did not wear the Myo armband correctly or if there was a new subject, the gesture recognition accuracy significantly decreased.

The simplest solution to this problem was to ask subjects to participate in the experiment many times, but this would be time-consuming. In addition, it could not be guaranteed that the subjects executed gestures in the same way each time [28]. Therefore, it was challenging to train a robust model with a large quantity of data. To reduce the burden of data collection and the variability in data across subjects, TL was used. Note that even if TL is used to fine-tune the network model for new users, gesture recognition accuracy will still decrease if there is electrode displacement when the user rewears the Myo armband. It was evident that re-fine-tuning the model in such cases would be meaningless and time-consuming. Therefore, after each rewearing of the Myo armband by a user, an electrode displacement algorithm [29] was applied to determine the channel with the highest sEMG energy generated during the execution of the fifth gesture. This channel was considered the reference channel. Then, the order of the eight channels was revised based on the relative

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positions of each channel with respect to the reference channel to mitigate the effects of positional displacement.

The parameter transfer approach supposes parametric models in the source and target domains. The transferred knowledge is encoded into parameters and transferred from the source domain to the target domain. In recent years, the parameter transfer approach has been widely used in transferring network weights of a CNN network trained on a source domain. Among TL-based approaches, fine-tuning [30] and adaptive [31] approaches have been the most commonly used. The fine-tuning technique was employed in the present study. Specifically, we fine-tuned the fully connected layers, Dropout layers, and McDropout layers of the source domain model. Since the IMU signals are natural signals and are not affected by the cross-position and cross-user issues observed in sEMG signals, only the EMG_Network and Fusion_Network branches were fine-tuned to address overfitting. The steps in the proposed TL strategy are given below.

1) The source domain network mentioned in Section II-C was trained using the source dataset, resulting in a user-general model. The user-general model served as a universal network model for interuser gesture recognition.
2) As shown in Fig. 3, model fine-tuning was achieved by freezing all parameters except for the fully connected layers, Dropout, and McDropout in the EMG_Network and Fusion_Network branches. Since the data volume decreases when transferring the network model to a new user, it is necessary to increase the Dropout and McDropout rates to prevent overfitting. After parameter tuning, the optimal solution was found to be increasing the rates from 0.1 to 0.2. Subsequently, the weights of the fully connected layers were retrained, as indicated by the green shaded area in Fig. 3. Since the IMU signal is a physical signal, no fine-tuning was performed on the IMU_Network branch.
3) The network architecture of the pretrained user-general model was transferred to the target network, and the network model was fine-tuned using the target dataset. This process involved transferring the knowledge from the source domain to the target domain, resulting in a user-transfer model. The user-transfer model represents the network model specific to a new subject.

III. Experimental Results and Discussion

The proposed source domain multistream CNN was trained on the source dataset. The learning rate was initialized at 0.0104709, and the Dropout and McDropout rates were set to 0.1. Fourfold cross-validation was used in the training phase. Afterward, the user-general model was evaluated on the target dataset (i.e., all repetition trials of the six individuals were used as the testing set). After a thorough evaluation, the user-general source domain model with optimal parameters was saved. In the subsequent phase, this source domain model was retrained and evaluated by applying the proposed TL strategy. In this experiment, the leave-one-out method was used to divide the data into training and testing sets. This approach used all but one trial for fine-tuning while evaluation was conducted on the one left-out trial. This was repeated for all trials, and then we calculated the average accuracy. Meanwhile, the Dropout and McDropout rates were adjusted to 0.2. For each subject in the target dataset, five trials were fed to the source domain model to retrain the model using fine-tuning techniques, as presented in Section II-C; moreover, ten-fold cross-validation was used in the transferring phase. Thus, the training dataset consisting of five trials was divided into ten equal parts. Each part was sequentially selected as a validation set, while the remaining parts served as the training set. For each subject, multiple target models called user-transfer models were generated using ten-fold cross-validation. These models were then evaluated on the testing set, and the model with the highest accuracy was saved for further analysis. It is worth noting that, for a more accurate evaluation of the network model's performance, the above experiments were conducted 20 times.

In this section, we first analyzed different feature extraction methods (i.e., RAW versus CWT versus STFT) for sEMG signals, and it was discovered that the CWT was more suitable for extracting time-frequency domain features from sEMG signals. Following that, we compared the performance of the proposed model in cross-user gesture recognition using single signals and multistream fusion signals. Moreover, the performance of the proposed multistream CNN framework with and without TL was evaluated. Furthermore, the proposed approach was compared with previous methods. All these experiments were conducted on the self-collected XDMyo dataset.

A. Proposed Network Performance for Different Feature Sets

In recent years, several methods for recognizing gestures using deep learning have been proposed. These methods have shifted the related research from feature engineering to feature learning. Regardless of the type of applied approach, the goal is to improve the accuracy and robustness of a gesture recognition classifier. A previous work [16] demonstrated that the CWT-based ConvNet can achieve a recognition accuracy of 98.31% for seven gestures. Furthermore, Li et al. [32] performed a STFT on the sEMG data to extract spectrograms and proposed a multichannel CNN network. Their results demonstrated that classification accuracy could be enhanced by combining the time- and frequency-domain information using the time-frequency domain data of the signal spectrum. Inspired by these previous works, we considered that combining deep learning and feature engineering could enhance recognition accuracy.

The proposed CNN network was trained on the fusion of IMU and three different types of sEMG data separately to evaluate the performance of the proposed framework when inputting different feature sets. The three different types of sEMG data were raw sEMG data, data transformed by the STFT, and data transformed by the CWT. The user-general and user-transfer models were trained and tested on the XDMyo dataset, and comparison results are shown in Fig. 4. We can see that the recognition accuracy of the proposed network with the user-general model when inputting the combination of IMU and sEMG data transformed by the CWT was 89.99%. This accuracy was higher than that for raw data and

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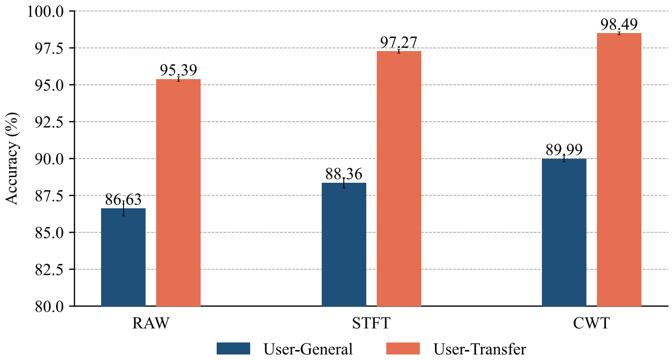

Fig. 4. Gesture recognition accuracy of the proposed method for different feature sets. The terms "user-general" and "user-transfer," respectively, represent the gesture classification accuracies of the source domain model and target domain model on the target dataset.

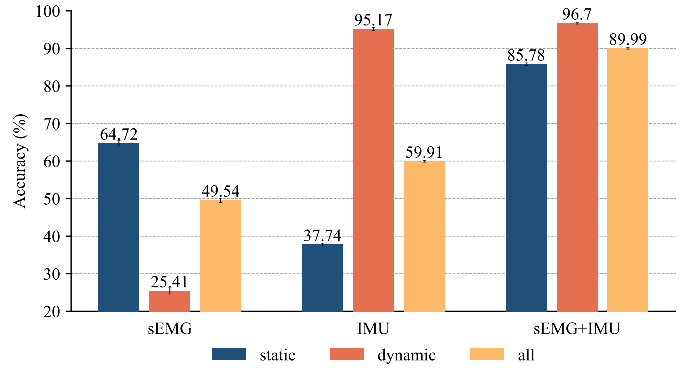

Fig. 5. Comparison of gesture recognition accuracy on a cross-user dataset using the proposed method for different input signals. "EMG" represents the use of only sEMG signals and the EMG_Network branch, while "IMU" represents the use of only IMU signals and the IMU_Network branch. "static" refers to the first seven static gestures, while "dynamic" refers to the remaining six dynamic gestures. The results for both "static" and "dynamic" gestures were computed based on the confusion matrix generated from "all" gestures.

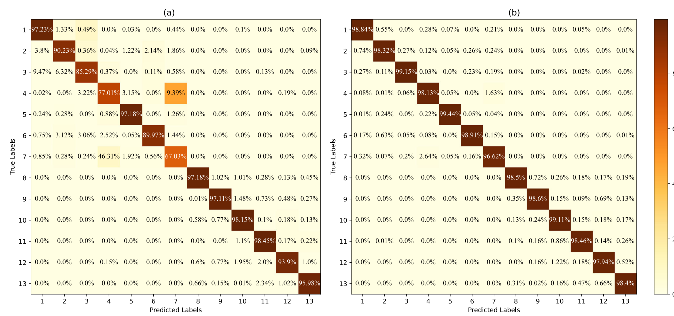

Fig. 6. Confusion matrices of (a) user-general and (b) user-transfer models. The diagonal elements represent "recall."

STFT-transformed data. The results also demonstrated that the average recognition accuracy of the proposed network with the user-transfer model when the IMU and CWT-based sEMG data were used as input was 98.49%. This accuracy was higher compared with those achieved when inputting raw sEMG data or data transformed by the STFT to the proposed network.

This experiment demonstrated that transforming the sEMG from the time domain to the time-frequency domain by the CWT is beneficial for improving the performance of the proposed CNN network. This is due to the limitations of the STFT, which cannot achieve high resolution in both frequency and time domains simultaneously. By contrast, the CWT overcomes this limitation by applying different scales and shifts on the time axis. Therefore, as described in Section II-D, the preprocessed IMU signals and sEMG signals transformed by the CWT were input into the proposed multistream CNN gesture recognition framework to achieve better gesture recognition performance.

B. Proposed Network Performance for sEMG and IMU Signals

The purpose of this experiment was to compare the performance of the proposed network for gesture recognition for single signal and multistream fusion signals. In this experiment, three signal combinations (i.e., sEMG signals, IMU signals, and the fusion of sEMG and IMU signals) were input in turn into the proposed framework without applying the TL strategy. The framework described in Section II-C was used for different signal combinations. For the sEMG signals, the IMU_Network and Fusion_Network branches (see Fig. 3) were abandoned; meanwhile, for IMU signals, only the IMU_Network branch (Fig. 3) was retained. In addition, a LogSoftmax activation function was applied to the output layer of EMG_Network and IMU_Network branches to obtain the probability distribution of gestures. When the fusion of sEMG and IMU signals was used as input data, the proposed framework was completely preserved.

The accuracy of the proposed framework was compared for static and dynamic gestures sensed by the sEMG and IMU signals both separately and simultaneously. The average accuracy of static gestures sensed by the sEMG data was 64.72%, while that of the IMU signals was 37.74%. However, the recognition accuracy of dynamic gestures sensed from the sEMG data was 25.41%, while that of IMU data was 95.17%. These results indicated that for static gestures, recognition performance was better when using the sEMG signals rather than the IMU signals; the opposite was true for dynamic gestures. As shown in Fig. 5, the best recognition performance for both static and dynamic gestures was achieved for the fusion of sEMG and IMU data. This observation suggested that there were variations in the sEMG signals for different dynamic gestures and variations in the IMU signals for different static gestures, thus indicating a complementary relationship between sEMG and IMU signals. To be precise, for all 13 gestures sensed from the combination of sEMG and IMU data, the average accuracy of the proposed framework without applying the TL strategy was 89.99%. Accuracy was significantly improved compared with the case when only sEMG or IMU data were used, which demonstrated that the fusion of signals could effectively capture the duality of dynamic and static gestures and be learned by the proposed network. Therefore, the fusion of sEMG and IMU data improved the recognition accuracy and reliability of both static and dynamic gestures.

C. Performance of Proposed Multistream CNN Framework With TL

The classification accuracy of the user-general and user-transfer models was compared using the XDMyo dataset to evaluate the performance of the proposed method. As presented in Fig. 6(a), for 13 gestures from new subjects, the

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average classification accuracy of the multistream CNN framework without TL was 89.99%. Moreover, the multistream CNN performed poorer in classifying static gestures than dynamic gestures. This indicates that IMU was a physical signal and did not vary with changes in subjects. Next, a fine-tuning transfer strategy was applied to EMG_Network and Fusion_Network branches of the proposed Multistream CNN framework. The user-general model was retrained on the training set of the target dataset, and the user-transfer model was obtained. Then, the retrained user-transfer model was tested on the testing set of the target dataset. As shown in Fig. 6(b), after applying the transfer strategy, the average classification accuracy increased to 98.49%.

The classification performance for all six subjects in the target dataset is presented in Fig. 7. The results show that the proposed transfer fine-tuning strategy significantly improved gesture recognition performance for all new subjects, particularly for subject S1. In addition, although there were some differences in accuracy among subjects before applying the TL strategy, the accuracy for all subjects remained in the same range after the transfer and recognition stability improved.

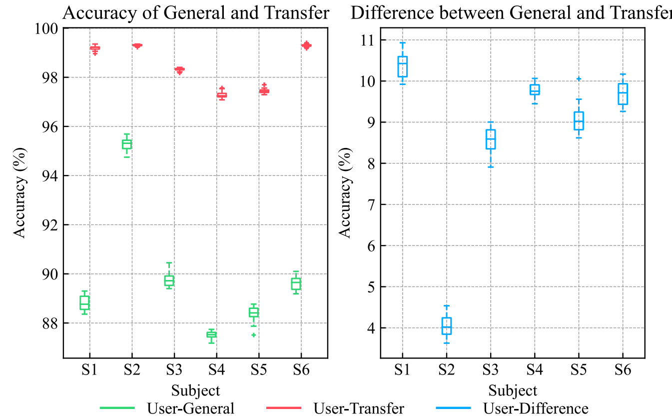

Fig. 7. Comparison of classification accuracy results of the proposed framework with and without TL for different subjects. The left plot represents the accuracy before and after TL for each participant, while the right plot shows the improvement in accuracy after transfer for each participant. The green line represents the accuracy of the user-general model, the red line represents the user-transfer model, and the blue line represents the difference in accuracy between the user-general and user-transfer models.

D. Comparison of the Proposed Nonhierarchical Approach With Previous Methods

To further analyze the advantages of nonhierarchicality and dual-modal fusion, other methods presented in relevant literature were tested on the XDMyo dataset, and their accuracies were compared with that of the proposed approach. Detailed comparison results are shown in Table I.

Chang et al. [9] employed a hierarchical approach for gesture recognition, in which a fourth-order autoregressive (AR) model and root mean square ratio (RMSR) were used to extract sEMG features, and energy was used to extract IMU features. Jiang et al. [8] extracted various temporal features from IMU and sEMG data and employed LDA for gesture recognition without using static–dynamic binary classification. Furthermore, Pan et al. [10] first extracted various temporal features from sEMG and accelerometer (Acc) data and then employed an SVM to distinguish between dynamic and static gestures. Finally, they used DBNs to further extract features and utilized an SVM for classification. Xu et al. [33] transformed the sEMG signals into energy spectrograms and fused them with IMU data, and then they employed a CNN to recognize gestures. Finally, Côté-Allard et al. [16] employed a CWT to extract time-frequency domain features from sEMG signals; then, they utilized a slow fusion and TL strategy to recognize static gestures. Since the datasets used in [16] and [33] did not include dynamic gestures and their recognition rates for the dynamic gesture dataset in the present article were low, only the recognition rates for static gestures are compared here.

In conclusion, the proposed method achieved better accuracy on the XDMyo dataset compared with the other methods. This could be due to the sEMG data measuring muscle activity while IMU data capturing movement information and that our proposed multistream CNN network takes advantage of both sEMG and IMU signals. It should be noted that, in the present study, the number of dynamic gestures was greater than in the previous studies listed in Table I. Based on these comparison results, we find that the proposed method can recognize both static and dynamic gestures directly rather than hierarchically.

E. Discussion

This article proposes a multistream CNN with a fine-tuning transfer framework for HGR. Compared with other methods, which train two classifiers to recognize static and dynamic gestures separately, the proposed framework can be used to recognize static and dynamic gestures in a nonhierarchical way. Experimental results demonstrate the feasibility of the fusion of sEMG and IMU data for the recognition of seven static and six dynamic gestures. Furthermore, these results show that TL can be used to improve the classification accuracy of gestures made by new subjects.

Although many sEMG-based gesture recognition methods have been proposed, the IMU-based gesture recognition methods have advantages in recognizing dynamic gestures. In this study, the proposed network performance for a single signal and multistream fusion signals is shown in Fig. 5. The results show using multistream fusion signals significantly improves accuracy compared with when only sEMG or IMU data are used. Therefore, the fusion of sEMG and IMU data can improve the recognition accuracy and reliability of both static and dynamic gestures, which may be related to the intrinsic correlations within the neurons of neural networks.

A detailed comparison of the performance of the proposed multistream CNN framework with and without a TL strategy is shown in Fig. 7. The proposed TL strategy significantly improves gesture recognition performance for all new subjects, particularly for subject S1, for which the classification accuracy improved by more than 10%.

A comparison of the proposed framework with those in previous studies is given in Table I. The proposed method achieved better accuracy than other methods. We propose a nonhierarchical multistream CNN network to take advantage of both sEMG and IMU signals, thereby improving the recognition rates of both static and dynamic gestures. Furthermore, compared with previous studies, the proposed method can recognize more dynamic gestures.

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TABLE I
COMPARISON OF OUR PROPOSED METHOD AND EXISTING METHODS ON THE XDMYO DATASET

	Chang [9]	Jiang [8]	Pan [10]	Xu [33]	Côté-Allard [16]	This work
Signal	sEMG+IMU	sEMG+IMU	sEMG+Acc	sEMG+IMU	sEMG	sEMG+IMU
Hierarchical	Yes	No	Yes	No	No	No
Dynamic gestures	Yes	Yes	Yes	No	No	Yes
Method	SVM	LDA	SVM+DBN	CNN+Energy Kernel	CNN+TL	CNN+TL
Accuracy	90.76%	83.45%	92.94%	83.83%	97.49%	98.5% / 98.49%

IV. VALIDATION FOR REAL-TIME CLASSIFICATION

Most gesture recognition research has only tested offline classification and has not evaluated recognition accuracy in real-time scenarios. Real-time recognition often requires dealing with more complex environments, and thus we must consider not only accuracy but also the algorithm's processing time to respond promptly to user gestures. Therefore, to assess the real-time performance of the proposed method in gesture recognition, we conducted online gesture recognition experiments. First, the preprocessing pipeline was modified to enable real-time recognition. Then, six subjects of the target dataset were selected as participants for the real-time experiment. These experiments were conducted using the best user-transfer model of each participant. The average recognition accuracy and processing time for each gesture and participant were then analyzed.

A. Experimental and Evaluation Metrics

We evaluated the real-time performance of the user-transfer models of each participant in the target dataset. Due to the standardization of acceleration for dynamic gestures in the preprocessing stage, we employed IMU threshold detection to distinguish between dynamic and static gestures before recognition. Then, preprocessing and feature extraction were performed in each sliding window; this was followed by feeding the processed data into the network for recognition. Since the signal was segmented using overlapping sliding windows, an initial accumulation of 56 samples was required, and this was followed by recognition every 20 samples (i.e., we employed a sampling rate of 200 ms, with a window size and stride of 280 and 100 ms, respectively). Hence, the initial accumulation of 56 samples was disregarded in this experiment. The experiment consisted of five rounds, with each round involving the recognition of 13 gestures. Each gesture was held for approximately 3 s, with a rest period of 5–10 s between each gesture. Participants were instructed to properly wear the Myo armband and regulate their movements based on visual feedback provided on an LED screen.

To evaluate the real-time recognition performance of the proposed network model, the recognition latency and positive predictive values (PPVs) [34] for each gesture of each subject in one trial were recorded. Here, latency refers to the total time elapsed from data acquisition to the completion of model recognition, including preprocessing and feature extraction. Tam et al. [34] noted that PPV offers a more nuanced assessment of accuracy and that evaluating time-distributed accuracy

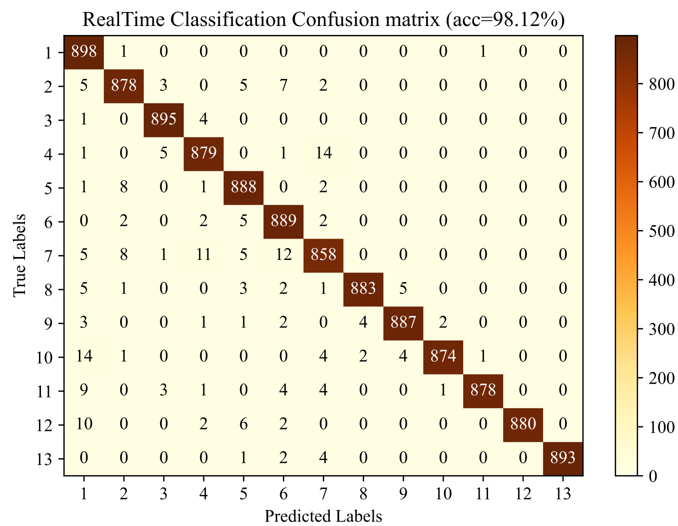

Fig. 8. Confusion matrices for real-time classification based on the user-transfer model.

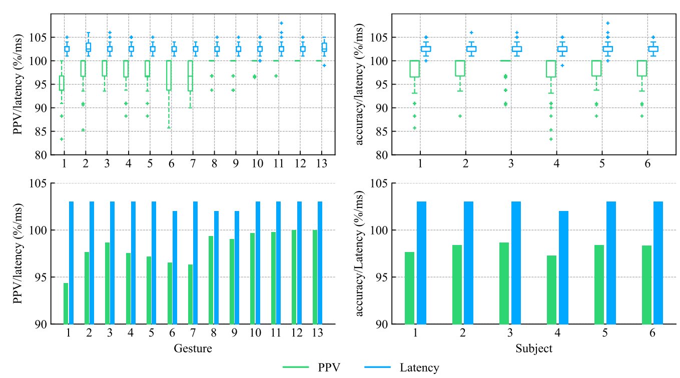

Fig. 9. PPV and latency with real-time classification. The color green denotes PPV, while the color blue denotes algorithm latency. The left half of the figure presents the distribution and mean values of PPV and latency data for each gesture, while the right half displays the overall accuracy and latency for each participant.

against a single target yields satisfactory results. Furthermore, the PPV metric allows us to better handle the classification results during transients, i.e., gesture transitions. Therefore, in our study, PPV was employed to assess the stability of online recognition.

B. Results and Discussion

Fig. 8 presents the number of correct/incorrect labels. The recall for each gesture is relatively high, except for the seventh gesture (i.e., the open hand). This can be attributed to slight wrist bending or tremors during gesture execution, which

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leads to a decreased recognition rate. In addition, there are instances where dynamic gestures are mistakenly recognized as static gestures, particularly the resting gesture. From Fig. 9, it can be observed that the resting gesture has the lowest PPV. Users tended to unintentionally pause or exhibit tremors during the execution of dynamic gestures, resulting in misclassification as the resting gesture. The right half of Fig. 9 reveals significant variations in online recognition accuracy among participants, which can be attributed to differences in forearm circumference, upper limb flexibility, and muscle development. The maximum delay for the sliding window is 100 ms, followed by an additional 5 ms approximately for window data preprocessing, feature extraction, and model recognition. Furthermore, switching between online recognition threads requires approximately 2–3 ms. In summary, the proposed method achieves an overall accuracy of 98.12% and an average latency of approximately 103 ms in real-time gesture recognition, thus indicating its suitability for real-time scenarios. However, to further enhance recognition accuracy rates, participants are advised to continually correct their gestures during the following visual feedback.

V. Conclusion

This article proposes a multistream CNN framework using a TL strategy and the nonhierarchical method. Due to the mutual enhancement of sEMG and IMU signals in recognizing dynamic and static gestures, the proposed framework exhibits significant advantages in recognizing gestures. In addition, the fine-tuning transfer strategy is used to improve the recognition accuracy of new subjects. Moreover, the XDMyo dataset consisting of seven static and six dynamic gestures is constructed and divided into a source dataset and a target dataset. The overall accuracy of our framework before and after TL is 89.99% and 98.49%, respectively. This 8.5% improvement is mainly attributed to the fine-tuning technique, which allows learning from the data of new subjects based on source domain knowledge. The proposed method enables rapid training of the target domain model for new users with few samples.

Different dynamic gestures exhibit various sEMG signals, while different static gestures generate subtle micromovements. Therefore, in this study, we employ a nonhierarchical multistream CNN network to learn the fused sEMG and IMU information, thereby significantly improving the overall classification accuracy of gestures. Hierarchical methods can only recognize either dynamic or static gestures, but our nonhierarchical method can simultaneously recognize combined gestures involving both dynamic and static components. For instance, these methods can be used to enable someone to manipulate a robotic arm by simultaneously clenching/unclenching their fists and swinging their arms. We demonstrate the advantages of nonhierarchicality and multistream fusion compared with using only sEMG or IMU for gesture recognition.

To achieve more natural HCI, it is necessary to incorporate more daily activity gestures (i.e., American Sign Language) and employ a nonhierarchical approach to classify combined gestures. Furthermore, the experimental results of this study indicate the complementarity of sEMG and IMU signals, which necessitates further investigation into the inherent neural correlations between sEMG and IMU in the multistream fusion network. Since the network architecture proposed in this study still exhibits overfitting, future works should explore the application of generative adversarial networks (GANs) from the computer vision domain to enhance digital signals.

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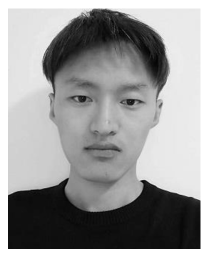

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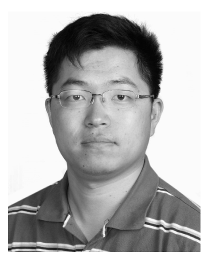

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